

Week 1: Issue Data and Vocabulary

Apache MapReduce - DSSE Assignment 3

1. Scope and completion

The complete list of 587 MapReduce issues from Issues.xlsx was processed. Issue records were downloaded from the Apache Jira REST API and cached locally. All assigned issues were retrieved successfully; no downloads failed.

Measure	Result
Assigned/downloaded issues	587 / 587
Issues with a parent	49
Processed tokens	35,666
Unique vocabulary terms	3,896
Document-term matrix	587 x 3896
Non-zero matrix entries	21,995

2. Jira data collection

For every issue ID, the pipeline requested all Jira fields and retained the complete raw fields inside the JSONL dataset. A compact CSV contains the most useful analysis fields: issue ID, three design-decision labels, parent ID, summary, type, status, priority, resolution, dates, comment counts, and attachment count. The parent field was absent for most issues, as expected.

The supplied bot-author list is built into the pipeline. The authoritative Jira total comment count is stored for each issue; human-comment counts are also calculated for the comments included in the issue response.

3. Text preprocessing

- Concatenated each issue summary and description.
- Converted text to lowercase and tokenized alphabetic/alphanumeric software terms.
- Removed English stop words, short tokens, numeric-only tokens, and URL-like tokens.
- Applied English Snowball stemming.
- Preserved the ordered token list for every issue in issues_processed.jsonl.

4. Vocabulary results

Rank	Token	Frequency	Interpretation
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Rank	Token	Frequency	Interpretation
1	job	850	Project-specific candidate
2	task	761	Project-specific candidate
3	reduc	585	Project-specific candidate
4	use	503	Generic candidate
5	file	469	Keep/review
6	map	422	Project-specific candidate
7	data	280	Keep/review
8	hadoop	257	Project-specific candidate
9	run	257	Keep/review
10	time	232	Keep/review
11	need	225	Generic candidate
12	schedul	225	Keep/review
13	current	208	Generic candidate
14	memori	194	Keep/review
15	user	194	Keep/review
16	implement	190	Generic candidate
17	output	190	Keep/review
18	support	190	Generic candidate
19	cluster	187	Keep/review
20	node	168	Keep/review

5. Candidates for refinement

The most frequent terms reveal several likely refinements before LDA. Project names and platform vocabulary (hadoop, mapreduce) may dominate topics without describing a design concern. Generic action words (use, need, current, new, implement, support) are also weak discriminators and are candidates for an expanded domain stop-word list. Terms such as job, task, mapper, reducer, JobTracker, TaskTracker, cluster, and node should not be deleted automatically: they can be mapped to a Component ontology class during the Week 2 experiment. Data/file/input/output may map to Data; shuffle and scheduling-related

terms should be retained initially because they can distinguish communication and resource-management topics.

6. Deliverables

- **week1_mapreduce.py** - Reproducible downloader and preprocessing pipeline
- **issues_processed.jsonl** - Full issue fields plus processed tokens
- **issues_metadata.csv** - Compact tabular issue metadata
- **vocabulary.csv** - Corpus and document frequencies
- **document_term_matrix.npz/.mtx** - Sparse DTM for LDA and other tools
- **document_ids.json / vocabulary_terms.json** - Matrix row and column mappings
- **summary.json / download_failures.json** - Validation summary and failure audit